



# **Artificial Intelligence (Machine Learning & Deep Learning) [Course]**

## **Week 3 - Descriptive Statistics - SeaBorn**

**[See examples / code in GitHub code repository]**

**It is not about Theory, it is 20% Theory and 80% Practical –  
Technical/Development/Programming [Mostly Python based]**

# Theoretical Background

## What Are Descriptive Statistics?

Descriptive statistics are brief informational coefficients that summarize a given dataset, which can be either a representation of the entire population or a sample of a population. Descriptive statistics are broken down into measures of central tendency and measures of variability (spread). Measures of central tendency include the mean, median, and mode, while measures of variability include standard deviation, variance, minimum and maximum variables.

## References:

[https://www.investopedia.com/terms/d/descriptive\\_statistics.asp](https://www.investopedia.com/terms/d/descriptive_statistics.asp)

<https://corporatefinanceinstitute.com/resources/data-science/descriptive-statistics/>



# Theoretical Background

## Probability

Probability is simply how likely something is to happen.

Probability means possibility. It is a branch of mathematics that deals with the occurrence of a random event.

Example 1:

There are 6 pillows in a bed, 3 are red, 2 are yellow and 1 is blue. What is the probability of picking a yellow pillow?

Ans: The probability is equal to the number of yellow pillows in the bed divided by the total number of pillows, i.e.  $2/6 = 1/3$ .

Example 2: Flipping a coin:

$$P(H) = \frac{1}{2} = 50\%$$

## References:

<https://www.khanacademy.org/math/statistics-probability/probability-library/basic-theoretical-probability/a/probability-the-basics>

<https://byjus.com/maths/probability/>

<https://www.cuemath.com/data/probability/>

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# Theoretical Background

## Data and its types (structured, Unstructured)

| Properties                 | Structured data                                                                                                                                             | Unstructured data                                                                                                                        |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Format examples            | <ul style="list-style-type: none"><li>• CSV</li><li>• Excel</li></ul>                                                                                       | <ul style="list-style-type: none"><li>• audio files (WAV, MP3, OGG)</li><li>• PDF documents</li><li>• images (JPEG, PNG, etc.)</li></ul> |
| Sources examples           | <ul style="list-style-type: none"><li>• online forms</li><li>• point-of-sale (POS) systems</li><li>• online transaction processing (OLTP) systems</li></ul> | <ul style="list-style-type: none"><li>• emails</li><li>• social media posts</li><li>• multimedia files</li><li>• IoT outputs</li></ul>   |
| Nature of data             | Quantitative                                                                                                                                                | Qualitative                                                                                                                              |
| Databases                  | Relational (SQL)                                                                                                                                            | Non-relational (NoSQL)                                                                                                                   |
| Storage for analytics use  | Warehouses and data lakehouses                                                                                                                              | Data lakes and data lakehouses                                                                                                           |
| Specialists to handle data | Business analysts, software engineers, data analysts                                                                                                        | Data scientists, data engineers, data analysts                                                                                           |
| Main benefits              | Easy to search and analyze, doesn't require much space                                                                                                      | Easy to collect and store                                                                                                                |
| Main challenges            | All data must fit predefined schema                                                                                                                         | Difficult to search and analyze                                                                                                          |

References: <https://www.altexsoft.com/blog/structured-unstructured-data>  
<https://www.ibm.com/think/topics/structured-vs-unstructured-data>



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# Measures of Central Tendency

A measure of central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data.

## Mean (Arithmetic)

The mean is equal to the sum of all the values in the data set divided by the number of values in the data set.

## Median

The middle score for a set of data that has been arranged in order of magnitude.

## Mode

The most frequent score in our data set.

### Example:

65 55 89 56 35 14 56 55 87 45 92

Mean: 59 , Median: 56 , Mode: 56 , 55

### References:

<https://statistics.laerd.com/statistical-guides/measures-central-tendency-mean-median.php>

<https://byjus.com/maths/central-tendency/>

<https://www.scribbr.com/statistics/central-tendency/>

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# Measures of Position

Measures of position give a range where a certain percentage of the data fall. The measures we consider here are percentiles and quartiles.

## Variance

The average squared deviation from the mean of the given data set

$$\sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n}$$

## Standard Deviation

The square root of the variance gives the "standard deviation"

$$\text{S.D.} = \sqrt{\text{Variance}} = \sigma$$

## Coefficient of Variation

The ratio of the standard deviation to the mean of the data set

$$(\text{S.D.} / \text{Mean}) * 100$$

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**References:** <https://online.stat.psu.edu/stat500/lesson/1/1.5/1.5.2>  
[https://stats.libretexts.org/Courses/Las\\_Positas\\_College/Math\\_40%3A\\_Statistics\\_a\\_Description/3.03%3A\\_Measures\\_of\\_Position](https://stats.libretexts.org/Courses/Las_Positas_College/Math_40%3A_Statistics_a_Description/3.03%3A_Measures_of_Position)  
<https://openstax.org/books/principles-data-science/pages/3-3-measures-of-position>



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# Measures of Dispersion

Measures of dispersion are non-negative real numbers that help to gauge the spread of data about a central value.

## Quartiles

Quartiles are numbers that separate the data into quarters.

first quartile is at position  $(n+1)/4$ , second quartile (i.e. the median) is at position  $2(n+1)/4$ , and the third quartile is at position  $3(n+1)/4$ .

## Percentiles

Percentiles provide a way to assess and compare the distribution of values and the position of a specific data point in relation to the entire dataset by indicating the percentage of data points that fall below it.

$$\text{Percentile} = \frac{\text{number of data values below the measurement}}{\text{total number of data values}} \times 100\% = \frac{n}{N} \times 100\%$$

## z-score

The z-score is a measure of the position of an entry in a dataset that makes use of the mean and standard deviation of the data.

$$z = \frac{x - \mu}{\sigma}$$

Where:

$x$  is the measurement

$\mu$  is the mean

$\sigma$  is the standard deviation

**References:** <https://online.stat.psu.edu/stat500/lesson/1/1.5/1.5.2>  
[https://stats.libretexts.org/Courses/Las\\_Positas\\_College/Math\\_40%3A\\_Statistics\\_and\\_Probability/Chapter\\_3/3.03%3A\\_Measures\\_of\\_Position](https://stats.libretexts.org/Courses/Las_Positas_College/Math_40%3A_Statistics_and_Probability/Chapter_3/3.03%3A_Measures_of_Position)  
<https://openstax.org/books/principles-data-science/pages/3-3-measures-of-position>



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Seaborn is a library that uses Matplotlib underneath to plot graphs. It will be used to visualize random distributions.

## Python Seaborn Module

- **Data visualization** is considered as the best way to depict and analyze the data
- Python Seaborn module basically serves the purpose of **Data Visualization** at an ease with higher efficiency.
- It supports **NumPy** and **Pandas** data structure to represent the data sets.
- Seaborn stands out to have a better set of functions to carry out data visualization than **Matplotlib** in an optimized and efficient manner.





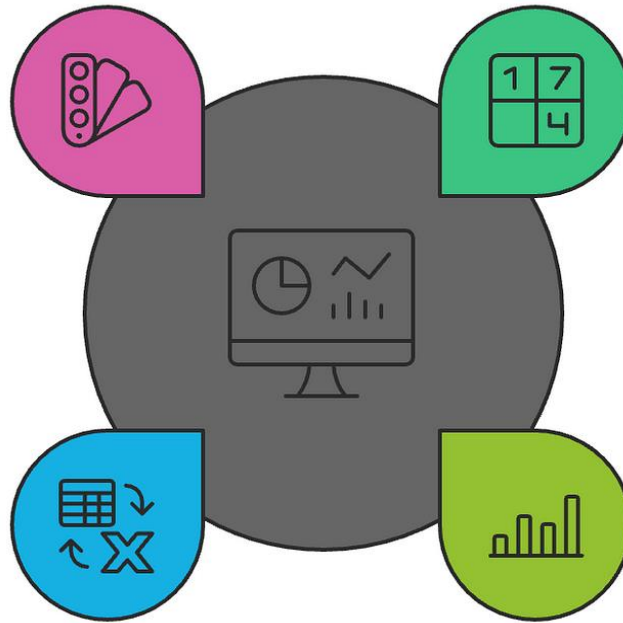
## Key Features of Seaborn

### Aesthetic Themes

Provides appealing themes and palettes

### DataFrame Handling

Directly works with data frames for analysis



### Simplified Plot Creation

Reduces code complexity for creating plots

### Statistical Visualization Focus

Emphasizes visualizing statistical data

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## MATPLOTLIB VS SEABORN



- 1 Can contain dissimilar data type.
- 2 Tabular operations, SQL like schemantics preprocessing task.
- 3 Two dimensions.
- 4 More memory.
- 5 Slower.



- 1 Has Homogeneous data.
- 2 Numeric computing, matrix & vector ops.
- 3 Multi-dimensional (>2 possible).
- 4 Less memory.
- 5 Faster.

### Reference:

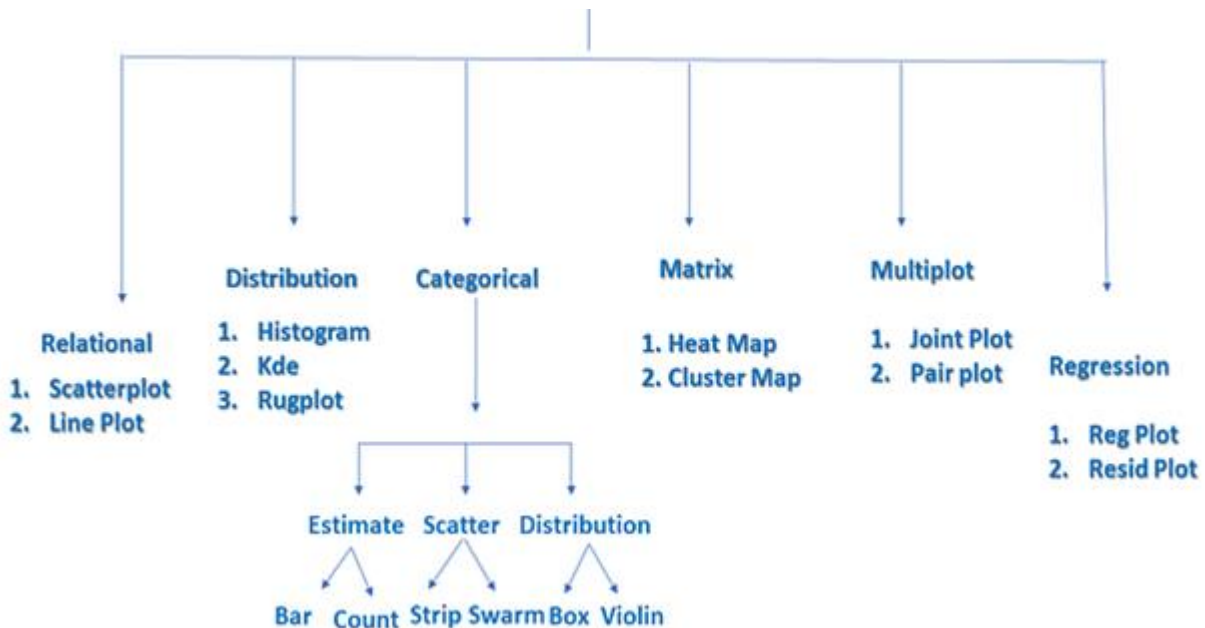
<https://docs.kanaries.net/topics/Seaborn/seaborn-vs-matplotlib>

<https://www.newhorizons.com/resources/blog/how-to-choose-between-seaborn-vs-matplotlib>

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## Reference:

<https://medium.com/womenintechology/complete-seaborn-tutorial-a5e184089c76>  
<https://towardsdatascience.com/14-data-visualization-plots-of-seaborn-14a7bdd16cd7/>  
<https://www.analyticsvidhya.com/blog/2021/12/12-data-plot-types-for-visualization/>

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## Exercises



## 3 Plotting With Seaborn

### Axis Grids

```
>>> g = sns.FacetGrid(titanic, #Subplot grid for plotting conditional relationships
                      col="survived",
                      row="sex")
>>> g = g.map(plt.hist, "age")
>>> sns.factorplot(x="pclass", #Draw a categorical plot onto a Facetgrid
                  y="survived",
                  hue="sex",
                  data=titanic)

>>> sns.lmplot(x="sepal_width", #Plot data and regression model fits across a FacetGrid
              y="sepal_length",
              hue="species",
              data=iris)

>>> h = sns.PairGrid(iris) #Subplot grid for plotting pairwise relationships
>>> h = h.map(plt.scatter)
>>> sns.pairplot(iris) #Plot pairwise bivariate distributions
>>> i = sns.JointGrid(x="x", #Grid for bivariate plot with marginal univariate plots
                    y="y",
                    data=data)

>>> i = i.plot(sns.regplot,
              sns.distplot)
>>> sns.jointplot("sepal_length", #Plot bivariate distribution
                 "sepal_width",
                 data=iris,
                 kind='kde')
```

### Regression Plots

```
>>> sns.regplot(x="sepal_width", #Plot data and a linear regression model fit
                y="sepal_length",
                data=iris,
                ax=ax)
```

### Distribution Plots

```
>>> plot = sns.distplot(data.y, #Plot univariate distribution
                        kde=False,
                        color="b")
```

### Matrix Plots

```
>>> sns.heatmap(uniform_data, vmin=0, vmax=1) #Heatmap
```

### Categorical Plots

#### Scatterplot

```
>>> sns.stripplot(x="species", #Scatterplot with one categorical variable
                  y="petal_length",
                  data=iris)
>>> sns.swarmplot(x="species", #Categorical scatterplot with non-overlapping points
                  y="petal_length",
                  data=iris)
```

#### Bar Chart

```
>>> sns.barplot(x="sex", #Show point estimates & confidence intervals with scatterplot glyphs
                y="survived",
                hue="class",
                data=titanic)
```

#### Count Plot

```
>>> sns.countplot(x="deck", #Show count of observations
                  data=titanic,
                  palette="Greens_d")
```

#### Point Plot

## 4 Further Customizations

Also see Matplotlib

### Axisgrid Objects

```
>>> g.despine(left=True) #Remove left spine
>>> g.set_ylabel("Survived") #Set the labels of the y-axis
>>> g.set_xticklabels(rotation=45) #Set the tick labels for x
>>> g.set_xlabel("Survived", #Set the axis labels
                "Sex")
>>> h.set(xlim=(0,5), #Set the limit and ticks of the x-and y-axis
        ylim=(0,5),
        xticks=[0,2.5,5],
```

## Python Seaborn Cheat Sheet

<https://www.datacamp.com/cheat-sheet/python-seaborn-cheat-sheet>

<https://cheatography.com/justin1209/cheat-sheets/seaborn/>

<https://book-of-gehn.github.io/articles/2021/06/05/Seaborn-Cheatsheet.html>

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See code here: <https://github.com/ShahzadSarwar10/ARTIFICIAL-INTELLIGENCE-SUMMER-OF-CODE-NAVTT-Sept2025/blob/main/Week3/Case3-3-Seaborn-Zameencom-property-data-by-Kaggle.py>

You should be able to analyze – each code statement, you should be able to see trace information – at each step of debugging. “DEBUGGING IS BEST STRATEGY TO LEARN A LANGUAGE.” So debug code files, line by line, analyze the values of variable – changing at each code statement. BEST STRATEGY TO LEARN DEEP.

Let's put best efforts.

Thanks.

Shahzad – Your AI – ML Instructor

25

## Exercises



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# Covariance vs Correlation

## Difference Between Correlation And Covariance

### Correlation

1. Mathematical concept used to measure the relationship between two variables
2. Shows the connection between the variables
3. Its value lies between -1 and +1
4. Not influenced by the change in the scale

[maindifferences.blogspot.com](http://maindifferences.blogspot.com)

### Covariance

1. Mathematical concept used to measure the variation between two variables
2. Shows the variability between the variables
3. Its value lies between  $-\infty$  and  $+\infty$
4. Easily influenced by the change in the scale

### References:

<https://www.coursera.org/in/articles/covariance-vs-correlation>

<https://builtin.com/data-science/covariance-vs-correlation>

<https://www.excelmojo.com/covariance-vs-correlation/>

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# Outlier Detection, and Data Interpretation

## References:

<https://www.geeksforgeeks.org/data-science/detect-and-remove-the-outliers-using-python/>

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Thank you - for listening and participating

- ☐ Questions / Queries
- ☐ Suggestions/Recommendation
- ☐ Ideas.....?

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